

TOFU: Trust-based Task Offloading in UAV-as-a-Service

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Abstract—This work proposes a trust evaluation scheme, TOFU, for task offloading in UAV-as-a-Service. Multiple UAV owners participate and collaboratively offer UaaS to the end-users. In the process, a UAV with limited resources offloads the task to other UAVs with adequate resources. However, not all the UAVs are trustworthy to offload the task. In such a case, we need to evaluate the trust of those UAVs to whom another UAV wishes to offload the task. The proposed trust evaluation scheme, TOFU, consists of three components – (i) residual energy, (ii) data transmission rate, and (iii) reputation of a UAV. The residual energy and data transmission are the pre-determined components retrieved directly. On the other hand, we need to compute the reputation of a UAV specific to the UaaS environment. We consider the resources available with a UAV to determine its initial reputation. Further, we apply the debit-credit system to compute the reputation value. When a UAV offloads a task, its reputation is debited from its total reputation. On the other hand, the reputation is credited to that UAV to which the task is offloaded. We apply a reputation transfer-based bargaining game while prioritizing the UAVs with higher capability. In particular, we adopt *Rubinstein bargaining* model to evaluate the trust of the UAVs and design an incentivized offloading mechanism. We use the concept of *Subgame Perfect Nash Equilibrium (SPNE)* to decide an optimal amount of reputation to be transferred between two UAVs. The game is modeled in a way that encourages UAVs with lower reputations to participate in the game by making them more patient so that they can negotiate for higher fractions of reputation. Simulation results show that TOFU reduces delay and energy consumption per task by up to 1.74% and 2%, respectively. Additionally, TOFU reduces task failure rate by approximately 9% as compared to the traditional approaches such as DeTTO, DPTO, and SOA, respectively.

Index Terms—Unmanned Aerial Vehicle, UAV-as-a-Service, Trust evaluation, Task offloading, Rubinstein bargaining game

1 INTRODUCTION

The demand for Internet of Things (IoT) applications is rapidly growing in various fields, including agriculture [1], communication and logistics in the military [2], and optimized traffic management in intelligent transportation systems [3]. Additionally, we witnessed the integration of Unmanned Aerial Vehicles (UAVs) with IoT architecture for

efficiently managing the services such as surveillance, environmental monitoring, and delivery of goods [4]–[6]. However, the applications of UAVs are non-trivial technologies for common individuals due to the lack of adequate skills in procuring, deploying, piloting, and maintaining them. Therefore, Yapp *et al.* [7] proposed a UAV service provision platform as UAV-as-a-Service (UaaS). Further, the detailed architecture of UaaS is proposed by Pathak *et al.* [8]. In a UaaS, a UAV serves multiple applications simultaneously while adopting the concept of virtualization. A UAV consists of multiple heterogeneous sensors which serve different applications. A UAV processes multiple tasks of the same or different applications. However, a single UAV may not be capable of serving all tasks of an application. In such a case, multiple UAVs collaboratively participate to execute all the tasks and complete the application. Additionally, a UAV may not have adequate resources to complete the assigned task. Therefore, such UAVs offload the task to other UAVs for executing the task. However, the UAVs to which a task is offloaded must be trustworthy ones. Therefore, we propose a trust evaluation scheme, TOFU, for the UaaS architecture that ensures tasks are offloaded to UAVs with significant trust. We apply Rubinstein's bargaining game to determine a UAV's trust while using the reputation transfer mechanism concept.

1.1 Motivation

In UaaS, multiple UAVs collaborate to deliver services for end-user applications. Typically, these UAVs are owned by different UAV owners who have a financial interest in serving an application. A UaaS platform consists of heterogeneous UAVs in terms of their resource availability. A UAV with insufficient resources may not be capable of simultaneously serving multiple resource-hungry tasks. In such scenarios, a resource-constrained UAV needs to offload the task to a UAV with adequate resources. However, not all available UAVs with adequate resources are trustworthy enough to offload the task. Therefore, it is essential to design a mechanism to evaluate the trustworthiness of a UAV before offloading the task to it. The concept of UaaS has recently evolved, and the trust evaluation scheme for UaaS is not available in the existing literature. Focusing on the necessity of designing a trust evaluation scheme for UaaS and considering the absence of such a mechanism in the literature, we motivate to architect the scheme, TOFU. The

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proposed scheme, TOFU, determines the trustworthiness of a UAV by using a debit-credit score approach.

1.2 Contribution

In this work, we proposed a scheme, TOFU, that evaluates the trust of the UAVs for task offloading for UaaS platform. The specific *contributions* of this work are as follows:

- To evaluate the trust of a UAV, reputation plays a crucial role. Therefore, we instrument an approach to determine the reputation by considering UaaS architecture. We consider the past reputation and the award earned by a UAV to compute the current reputation score. A debit-credit-based reputation transfer approach is used to calculate the award of a UAV. The proposed debit-credit-based approach is suitable for UaaS architecture while considering the presence of multiple heterogeneous UAVs.
- Deciding the award of a UAV in the UaaS platform is a non-trivial task. Therefore, we develop the mathematical model for award computation using Rubinstein's bargaining game. In particular, we determine an optimal award score by implementing Subgame Perfect Nash Equilibrium (SPNE).
- The proposed scheme, TOFU, is new and specifically tailored for UaaS. The performance evaluation is an essential to determine the applicability of the TOFU for UaaS. Therefore, we evaluate the performance of TOFU by conducting rigorous simulations. We consider different performance metrics – energy consumption, delay, reputation, failure rate, and computational capabilities – in our simulation. Additionally, we compare the performance of TOFU with the three closest possible existing approaches – DeTTO, DPTO and SOA.

2 RELATED WORK

In this section, we elaborately discuss the existing work related to our proposed work. We divided this section in the following parts:

2.1 Virtualization-based service provisioning architecture

Vendor lock-in and lack of service interoperability hinder the development, deployment, and scheduling of Backend-as-a-Service (BaaS) enabled serverless workflows in federated serverless infrastructures. To address this, Larcher et al. [9] proposed BaaSLess, a platform that delivers global and dynamic federated BaaS to serverless workflows. Further, to address demands due to the irregular and unpredictable nature of traffic from various applications for Satellite-as-a-Service (SataaS), Kebedew et al. [10] presented a dynamic capacity allocation framework that efficiently handles diverse traffic flows with varying arrival rates, aiming to minimize rental costs while meeting blocking probability and QoE requirements. Similar to traditional cloud provisioning, several works studied virtualization strategies for Sensor-as-a-Service (Se-aas) to improve trust, energy efficiency, and quality of service. Mondal et al. [11] proposed

VISE, a game-theoretic virtual sensor orchestration scheme that enables optimal collaboration among sensor-cloud service providers (SCSPs) and vehicle owners (VOs), improving data delivery, revenue sharing, and network lifetime through dynamic SCSP and node selection. Extending the concept of Se-aas into the dynamic Internet of Vehicles (IoV) environment, Abbes et al. [12] introduced an IoV-Cloud architecture for intelligent provisioning of vehicle virtual sensors. Their approach incorporates functional layers that enable scalable, on-demand virtualization and management of physical sensors in mobile vehicles, addressing cost-effectiveness and profitability for providers in volatile markets. Focusing on resource efficiency in virtualized wireless sensor networks (WSNs), particularly for delay-sensitive applications, Maleki et al. [13] developed a Dynamic Network Embedding (DNE) heuristic. This solution reduces energy consumption while satisfying latency and bandwidth requirements, achieving near-optimal performance and demonstrating scalability in large-scale deployments. Similar to the sensor service provisioning approach, Yapp et al. [7] proposed the concept of a UAV service platform known as UAV-as-a-Service (UaaS) in which end users avail services from UAVs. Finally, UaaS is theoretically modeled, and different components are introduced by Pathak et al. [8]. On the other hand, to provide seamless UAV services to the end-users, Roy et al. [14] proposed Serv-HU in the UaaS platform. Serv-HU addresses the issue of services provided by a service provider to a limited application area by proposing a service hand-off mechanism to another service provider. FU-Serve is proposed by Imandi et al. [15] to address the issues of data transmission delay for serving time-critical IoT applications using the UaaS platform. Further, to overcome the limitations of latency, energy waste, and suboptimal utilization of resources, Imandi et al. [16] proposed Con-Fog, a consensus-based Fog UAV selection. Con-Fog selected the best fog nodes based on the geo-distance, link quality, computational resources, and residual energy. Building on their earlier work, the authors [17] further enhanced the FU-Serve platform by integrating Federated Learning (FL) to address the computational limitations of fog nodes. The authors proposed FL-based models for on-device image classification, enabling the efficient deployment of deep learning in resource-constrained environments.

2.2 Task offloading in mobile environment

To address the issues of the growing demand for service requests in the Internet of Vehicles (IoV), Pratap et al. [18] introduced CoTEV, a cooperative task execution framework. When roadside units (RSUs) struggle to meet task deadlines, CoTEV integrates by intelligently coordinating with trusted vehicles to ensure timely and efficient completion. In another work, Dass et al. [19] worked on dependency-aware trustworthy task offloading problem (DeTTO) for vehicular IoT networks. The authors designed a greedy algorithm to obtain an optimal set of nodes for task offloading. Kaur et al. [20] formulated the task offloading problem in MEC as a classification problem to minimize energy consumption while guaranteeing the constrained latency requirement. They used supervised learning to decide whether to offload a task to provide offloading utility and QoE preservation

in devices with different energy. Zhang et al. [21] presented an optimal-latency offloading approach with roadside units (RSUs) and assistant vehicles by jointly optimizing resource allocation and frequency reuse. A two-step optimization with particle swarm optimization (PSO) and iterative method is used to solve the mixed-integer non-linear programming (MINLP) problem efficiently. In order to improve the performance of the Road-Side Unit (RSU), Khabbaz [22] proposed a Deadline-Constrained Task Offload (R2V-DCTO) scheduling scheme. Similar to the vehicular networks, different authors addressed the issues of task offloading for UAV networks. Pamuklu *et al.* [23] designed a multi-actor-based risk-sensitive reinforcement learning approach for Aerial base stations (ABSs) task offloading while focusing on the agricultural scenario. Additionally, the authors presented a mixed integer linear programming solution to determine a lower bound on the performance. In another work, Alam *et al.* [24] proposed a task offloading scheme for Air-Ground Integrated Networks. With an objective to reduce the overall energy consumption of both users and UAVs, Poursiami *et al.* [25] proposed a task offloading mechanism. The authors examined a multi-UAV assisted Mobile Edge Computing (MEC) heterogeneous network composed of two types of UAVs – edge UAVs with computing resources and relay UAVs that facilitate users in sending tasks to Ground Base Stations. Along the same direction, El-Emary *et al.* [26] designed a task offloading approach for a UAV-assisted-MEC system while ensuring the minimization of the user equipment (UE) energy consumption.

Synthesis: Existing literature presented various service provisioning paradigms, including BaaSLess, SaaS, Se-aas, and UaaS, each addressing unique challenges in their respective domains. Additionally, the prior works examined task offloading strategies to enhance system performance, particularly in mobile environments such as vehicular networks and UAV networks. While several proposed schemes focuses on the trustworthy and efficient offloading, while considering the general edge computing or vehicular contexts. The literature highlights a research gap in trust-aware task offloading tailored specifically to UAV-as-a-Service (UaaS) scenarios, where UAVs themselves act as mobile service providers with limited and dynamic resources.

3 SYSTEM MODEL

Let a set of UAVs $U = \{U_1, U_2, U_3 \dots U_n\}$ be present in a UaaS architecture. The set of UAVs is divided into two disjoint subsets as U^T and U^S , representing task and service UAVs, respectively, such that $U^T \cup U^S \subseteq U$. In UaaS architecture, a few UAVs are idle, to which no task is allocated or offloaded. If a UAV is incapable of serving multiple tasks due to inadequate resources, the UAV offloads the task to a service UAV. Let $\mathbb{T} = \{\mathbb{T}_1, \mathbb{T}_2, \mathbb{T}_3, \dots, \mathbb{T}_k\}$ be the set of task to be assigned, and k is the total number of tasks. The i^{th} task constitutes two tuples, such that $\mathbb{T}_i = \langle t_i, U_i \rangle$, where t_i and U_i denote the task and the UAV it is assigned to, respectively. Therefore, we consider the presence of two types of UAVs – (i) task UAV - which offloads the task, and (ii) service UAV - which executes the offloaded task. In a collaborative UaaS setup, a task UAV needs to select a service UAV on which task can be executed reliably. Since

UAVs differ in energy, capability, and reliability, randomly selecting a UAV and offloading tasks on it may result in delays or failures in task execution. Therefore, we introduce TOFU – a trust evaluation scheme that helps task UAVs make smarter offloading decisions by assessing each service UAV's residual energy, data rate, and reputation. Such a mechanism ensures tasks are assigned to reliable UAVs, improving overall system performance and stability.

We consider the offloading of dynamic tasks in the proposed UaaS architecture, where each task $\mathbb{T}_i = \langle t_i, U_i \rangle$ may differ in its computational requirements and deadline constraints. Such a variability in computational requirements and deadline constraints reflects real-world applications, where tasks originating from different UAVs may demand varying levels of processing power and urgency. To address the issues of offloading of dynamic task, our model evaluates each task t_i based on its resource intensity and time sensitivity, thus ensuring that task UAVs U^T offload tasks to suitable service UAVs U^S capable of meeting those specific demands. Taking into account dynamic task profiles in the trust and eligibility assessment, TOFU enhances system responsiveness and task completion reliability.

In the proposed system, a UAV is associated with a trust score dependent on its residual energy, reputation, and data rate. We initialize all the UAVs with a reputation score based on the available computational resources in the UAV. Therefore, the change in the reputation scores impacts the trust in a UAV. We design a dynamic trust computation mechanism using a *bargaining game*, where the task and service UAVs negotiate the award.

TABLE 1: Key Notations

Symbol	Description
U	Set of All the UAVs
U_i^T	i^{th} UAV assigned with a task
U_j^S	j^{th} UAV to which task is offloaded
ζ_{ij}	Trust between U_i^T and U_j^S
$\mathcal{E}_j$	Residual energy of U_j^S
$\mathcal{R}_i$	Reputation score of U_i
$\mathcal{D}_{ij}$	Data transfer rate between U_i^T and U_j^S
η_t	Parameter to decide if a task t is successfully executed
T_j^t	Time taken in executing task t by U_j^S
ET^t	Maximum allowed delay for task t
$\mathcal{A}_{ij}^t$	Award offered by U_i^T to U_j^S for completion of task t

Definition 3.1. Trust in the context of task UAV and service UAV is defined as the degree of confidence a task UAV has in the ability of a service UAV to complete the offloaded task successfully within the specified constraints.

The trust between a service UAV, U_j^S and a task UAV, U_i^T is computed as a function of the reputation score, $\mathcal{R}_j$, residual energy, $\mathcal{E}_j$ of U_j^S and the data transmission rate, $\mathcal{D}_{ij}$, of the channel between U_j^S and U_i^T .

$$\zeta_{ij} = \mathcal{E}_j \times (r \cdot \mathcal{R}_j + d \cdot \sigma(\mathcal{D}_{ij})) \quad (1)$$

We scale $\mathcal{D}_{ij}$ to the range $[0, 1]$ using the sigmoid function σ and $(r, d) \in [0, 1]$. The initial trust value ζ_{ij} is bootstrapped

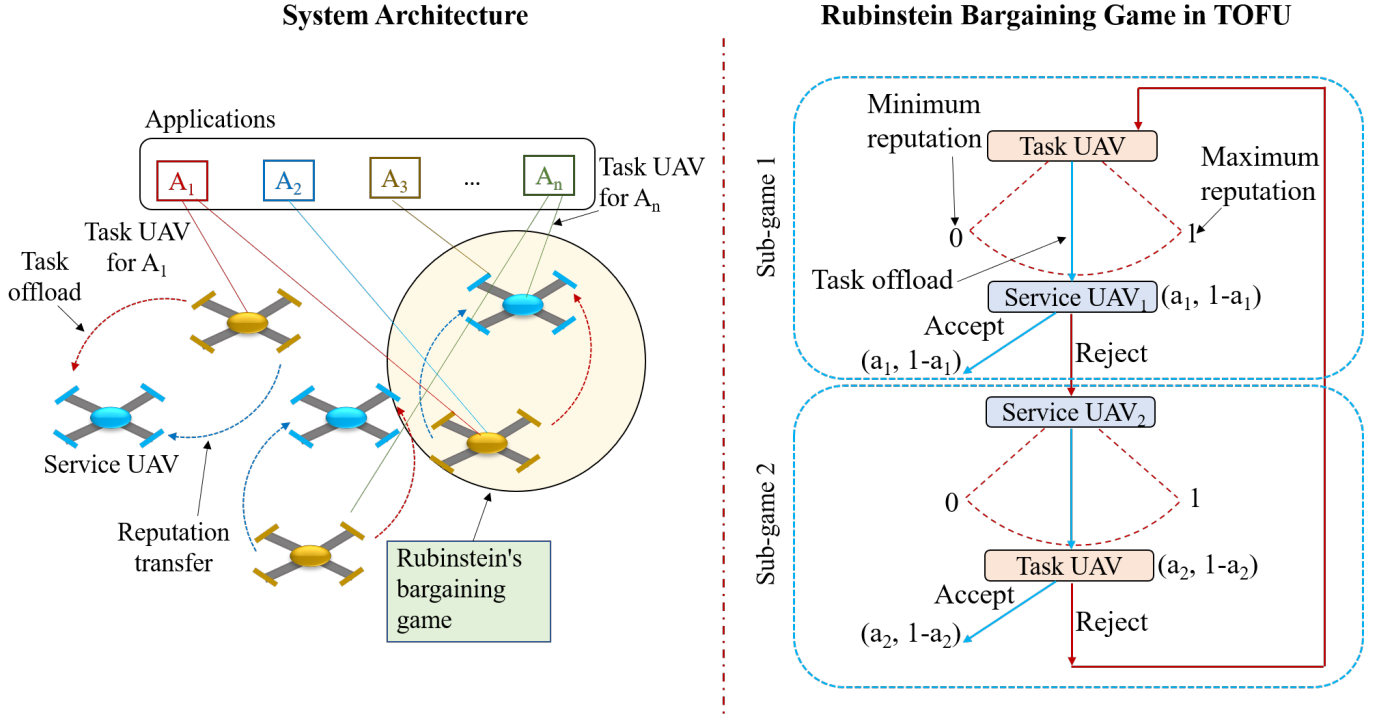


Fig. 1: Bargaining game model for task and service UAV.

using the residual energy $\mathcal{E}_j$, the data transfer rate $\mathcal{D}_{ij}$, and the reputation score $\mathcal{R}_j$. Since trust cannot be directly observed, we assume a neutral baseline trust value when a UAV U_i^T offloads a task to a UAV U_j^S with no prior interactions. This baseline is then adjusted dynamically based on UAV U_j^S 's performance in task execution, successful data transfers, and energy availability, ensuring a fair and adaptive trust evaluation over time.

Constraint 1. A task UAV, U_i^T , offloads a task, t to a service UAV, U_j^S if and only if it is eligible in executing the task, and the eligibility of U_j^S is defined as:

$$e(U_j^S, t, U_i^T) = \begin{cases} 1, & \text{if } \zeta_{ij} \geq \zeta_t^{\min} \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

where $\zeta_t^{\min}$ is the least required trust to offload a task.

For a given task, there is a maximum tolerable processing delay, memory required, number of instructions, and minimum data rate. Using these factors, we compute the minimum computational frequency required, f_t , as:

$$f_t = \frac{N_t}{T_t^{\text{proc}}} \quad (3)$$

N_t and T_t^{proc} indicate the number of instructions and the maximum tolerable processing delay in a task t , respectively. A task t offloaded to service UAV, U_j^S is considered successfully executed only if the execution time satisfies $T_j^t \leq ET^t$, where T_j^t denotes the time taken by U_j^S to execute the task and ET^t is the maximum allowed delay for task t .

Further, we compute $\zeta_t^{\min}$ as:

$$\zeta_t^{\min} = r \cdot (\tau \times (f_t \cdot M_t)) + d \cdot (\mathcal{D}_t^{\min}) \quad (4)$$

where M_t and $\mathcal{D}_t^{\min}$ are the memory and data rate requirements for a task, t .

The offload decision represents whether to offload the task. Further, the decision will be true if and only if the

required computational frequency and memory for the task are less than the available frequency and memory, respectively.

Constraint 2. At any given instant, there exists at least one UAV that is eligible to perform task t .

$$\exists U_j^S \in U^S \text{ s.t. } e(U_j^S, t, U_i^T) = 1 \quad (5)$$

If no UAV possesses the resources that are required to execute a given task, the task is considered infeasible and is bound to fail. In such scenarios, the proposed method does not force an ineligible UAV to attempt the task, as doing so will affect its reputation.

Constraint 3. The mapping from task UAVs to service UAVs is always one-to-one, which indicates that every task UAV offloads its task to one and only one service UAV.

$$\forall U_i^T \in U^T, \exists \text{ one and only } U_j^S \in U^S \quad (6)$$

Problem Statement:

In a traditional UaaS architecture, it is essential to offload tasks to ensure continuous and seamless service provisioning to end users. Efficient task offloading in UaaS architecture is a critical challenge. It involves transferring computational tasks from a task UAV to a service UAV while executing them within the stipulated time and ensuring the Quality-of-Service of the tasks. To achieve successful offloading of tasks, a task UAV must obtain a suitable service UAV that is capable of successfully executing the task within the specified specific time. Ensuring trust in the process is essential to maintain service quality. This leads to a trust evaluation scheme that quantifies each UAV's trust in every other UAV. Thus, in this work, we propose a dynamic trust evaluation scheme employing a fixed discounting factor bargaining game. Fig. 1 represents the system architecture of the UaaS, where a task UAV offloads the task to a service UAV. An application A_i is

associated with a task UAV. As task UAV is incapable of serving the task due to the unavailability of the resources, it finds a service UAV to complete the task. However, the task UAV transfers a fraction of its reputation as an award to a service UAV for completing the task. To offload the task and transfer its reputation, the task and service UAVs participate in Rubinstein's bargaining game. In Fig. 1, task UAV offers its fraction of reputation $(1 - a_1)$ to service UAV and keeps a_1 as its own reputation. The service UAV can either accept this offer or respond with a counteroffer, proposing to keep a larger share of the reputation $(1 - a_2)$ while offering a_2 to the task UAV, such that $(1 - a_2) > (1 - a_1)$ and $a_2 < a_1$. Based on the bargaining, if the task UAV is satisfied with a_2 , it offloads the task to the service UAV. Alternatively, the task UAV tries to bargain again with the same or another service UAV.

4 PROPOSED METHODOLOGY

We divide the methodology into two components — (i) *trust evaluation*, which is performed by every task UAV to calculate the trust score of all the service UAVs, and (ii) *dynamic score transfer*, which is performed using a two-player bargaining game between the task UAV and its associated service UAV. The two components are discussed in the following subsections:

4.1 Trust Evaluation

As mentioned in Section 3, the trust evaluation depends on three factors – *residual energy*, *reputation score* and *data transfer rate*. We use Peer to Peer reputation score, motivated by the work of Gupta *et al.* [27], to calculate the reputation, $\mathcal{R}_j$ of each of the service UAV, U_j^S , $\forall j \in \{1, 2, \dots, n(U^S)\}$. Then, we compute the maximum data transmission rate, D_{ij} , using Shanon's channel capacity [28] and a weighted sum for the final trust evaluation.

4.1.1 Residual Energy

The residual energy of a service UAV U_j^S , represented as $\mathcal{E}_j$ is a critical factor in evaluating its trustworthiness. A UAV with higher residual energy is more likely to sustain the energy-intensive processes involved in task execution and data transmission. Therefore, in our trust evaluation scheme, we incorporate energy to prioritize UAVs with higher residual energy to ensure reliability and reduce the risk of task failure.

4.1.2 Reputation Score

We use a debit-credit reputation computation mentioned in [27], where offloading a task or the unsuccessful execution of an offloaded task is considered a debit from task UAV. Successfully executing an offloaded task is considered a credit to the service UAV.

Definition 4.1. The Reputation score ($\mathcal{R}$) of a UAV is defined as a weighted sum of the current reputation score and debit-credit score available.

Using the above statement, we mathematically represent $\mathcal{R}_j$ of U_j^S after completing a task t from U_i^T as:

$$\mathcal{R}_j^t = (\mathcal{A}_{ij}^t \cdot \mathcal{R}_i^{t-1} - \eta_t) + \mathcal{R}_j^{t-1} \quad (7)$$

Similarly, the reputation score of U_i^T is computed as:

$$\mathcal{R}_i^t = \mathcal{R}_i^{t-1}(1 - \mathcal{A}_{ij}^t) \quad (8)$$

where η_t is a boolean value that indicates the truth of unsuccessful execution and is expressed as,

$$\eta_t = \begin{cases} 1, & \text{if } t \text{ is failure} \\ 0, & \text{otherwise} \end{cases} \quad (9)$$

and $\mathcal{A}_{ij}^t$, $\mathcal{R}_j$ and $\mathcal{R}_i$ are the award, current reputation score of the service UAV performing the task, and task UAV offloading the task, respectively.

Definition 4.2. Award, $\mathcal{A}_{ij}^t$, is the fraction of reputation score transferred from U_i^T to U_j^S for completing a task t .

The resources available at each UAV play an important role in computing the initial reputation. For instance, a UAV with a higher computational clock speed should have a higher reputation score than other UAVs with a lower clock speed value.

$$\mathcal{R}_j^0 = \tau \times (f_j \cdot M_j) \quad (10)$$

where f_j is the clock speed of U_j^S and τ is the time constant. τ is defined as the time taken to execute a task with the number of instructions per unit of data equal to the clock speed of the CPU; therefore, $\tau = 1s/bits$.

Property 1. The total reputation of the system, expressed as sum of the reputation scores of each UAV, is always conserved.

Proof. Consider there are k tasks assigned to k UAVs, i.e. one task is assigned to each UAV. The mapping from task UAVs to service UAVs is one-to-one, as mentioned in Constraint 3, and there exist k UAVs to which the task is offloaded. Randomly pick a pair, p , of UAVs $U_i^{k'}$ and $U_j^{k'}$ such that one is the task UAV, and another is the service UAV for a task k' .

Before task execution: Each UAV has a reputation score in the system at any time. Before executing task k' , the reputations of the task and service UAV are $\mathcal{R}_i$ and $\mathcal{R}_j$, respectively. Therefore, the total reputation of the pair is:

$$\mathcal{R}_p = \mathcal{R}_i + \mathcal{R}_j \quad (11)$$

After task execution: After the task is successfully offloaded from $U_i^{k'}$ to $U_j^{k'}$ and executed, the task UAV transfers a fraction of its reputation. From Eqs. (7) and (8), the updated reputations of the UAVs is expressed as:

$$\mathcal{R}_i^{k'} = \mathcal{A}_{ij} \times (\mathcal{R}_j + \mathcal{R}_i) \quad (12a)$$

$$\mathcal{R}_j^{k'} = \mathcal{R}_j \times (1 - \mathcal{A}_{ij}) \quad (12b)$$

Adding Eqs. (12a) and (12b), we obtain,

$$\mathcal{R}_p^{k'} = \mathcal{R}_i + \mathcal{R}_j \quad (13)$$

Comparing Eqs. (11) and (13), we observe that both the initial and final total reputations of the pair remain constant.

The total system reputation remains constant because the total reputation of any pair stays the same, and the reputations of UAVs not involved in task offloading do not change. □

4.1.3 Data Transmission Rate with Fading and Contention

The data transmission rate plays a crucial role in evaluating the trustworthiness of a service UAV, as higher data rates facilitate faster offloading and task execution. Traditionally,

we use Shannon's channel capacity to model the maximum data transmission rate:

$$\mathcal{D}_{ij} = \mathcal{B} \cdot \log_2(1 + SNR_{ij}) \quad (14)$$

Here, $\mathcal{B}$ is the bandwidth allocated for the communication link between task UAV U_i^T and service UAV U_j^S , and SNR_{ij} denotes the signal-to-noise ratio of the channel.

We model the SNR_{ij} considering Rayleigh fading [29] as:

$$SNR_{ij} = \frac{\rho_i \cdot |h_{ij}|^2}{N_0} \quad (15)$$

where ρ_i is the transmission power of U_i^T , N_0 is the noise power, and $|h_{ij}|^2$ denotes the channel power gain between the UAVs. We now consider small-scale Rayleigh fading and large-scale path loss:

$$|h_{ij}|^2 = \frac{z_{ij}}{d_{ij}^\gamma}, \quad z_{ij} \sim \text{Exp}(1) \quad (16)$$

where d_{ij} is the distance between UAVs and γ is the path loss exponent. The variable z_{ij} follows an exponential distribution (Rayleigh fading squared).

Channel Contention: In real-world deployments, a service UAV receive tasks from multiple task UAVs simultaneously. If n_j task UAVs are offloading to the same service UAV U_j^S , the effective bandwidth allocated to each communication link becomes $\mathcal{B}_{eff} = \frac{\mathcal{B}}{n_j}$. Thus, the adjusted data rate becomes:

$$\mathcal{D}_{ij}^{eff} = \frac{\mathcal{B}}{n_j} \cdot \log_2 \left(1 + \frac{\rho_i z_{ij}}{N_0 d_{ij}^\gamma} \right) \quad (17)$$

where n_j is updated dynamically during task assignment and reflects the number of active offloading links to U_j^S . Each service UAV manages contention locally by maintaining and updating its own n_j value.

4.2 Bargaining Game-Based Dynamic Score Transfer

We use a two-player bargaining game for dynamic score transfer with task and service UAVs as the players. The reputation score is a transferable quantity. If a UAV, U_i^T , offloading a task t to U_j^S , it must transfer a fraction of its score to the other UAV as the award, $\mathcal{A}_{ij}^t$, for executing the task offloaded. This directs to a trade-off between the task UAV and service UAV.

4.2.1 Justification for Bargaining Game

In the proposed task offloading scheme for UAVs, employing the *Rubinstein bargaining model* [30] to determine the reputation score transfer between task UAVs and service UAVs that ensures a fair and strategic trust distribution. The *Rubinstein model*, renowned for its ability to capture the dynamics of alternating offers and time preferences, is particularly suited for our application. By leveraging this model, we account for both parties' bargaining power and discount factors while ensuring that the fraction of the reputation score transferred reflects the value of successful task execution from the perspectives of both the task and service UAVs. This approach not only incentivizes service UAVs to perform reliably but also dynamically adjusts the reputation distribution based on actual performance, thereby enhancing the robustness and trustworthiness of the UAV network. Integrating the Rubinstein bargaining model into our trust evaluation scheme aligns with the principles of game theory, providing a mathematically sound and practical solution for

maintaining and fostering collaboration among UAVs in the network.

4.2.2 Game Formulation

We employ the Rubinstein bargaining model [30] with fixed discounting factors for the defined problem. We define the discount factor δ_i for task UAV U_i^T and δ_j for service UAV U_j^S based on their reputation scores. The discount factor is inversely related to the UAV's reputation, encouraging UAVs with lower reputation scores to be more patient and gain higher rewards for executing offloaded tasks. Since the bargaining takes place only among eligible UAVs, this design promotes participation from lower-reputation UAVs that meet the minimum resource criteria. The discounting factors are defined as follows:

$$\delta_i^T = \frac{\sum_k \mathcal{R}_k^0}{\mathcal{R}_i^T} \quad (18a)$$

$$\delta_j^S = \frac{\sum_k \mathcal{R}_k^0}{\mathcal{R}_j^S} \quad (18b)$$

The reason for assigning higher patience to UAVs with lower reputations is to incentivize their participation. By being more patient, these UAVs are more likely to negotiate for and secure greater awards over time. After every successful execution of a task t , the award offered by U_i to U_j is computed through *Sub-game Perfect Nash Equilibrium (SPNE)* of the bargaining game played by them.

Theorem 1. *In the formulated UAV offloading game, a Sub-game Perfect Nash Equilibrium (SPNE) exists that ensures both the task and service UAV reach an optimal agreement on the fraction of the reputation score to be transferred.*

Proof. Fig. 1 depicts the architecture of TOFU in which employment of the Rubinstein bargaining game is used in task offloading in UaaS. We start our game with an offer made by task UAV, as it offloads the task. In the game, we consider two sub-games – in one, the task UAV, U_i^T , is offering S_1 while in the other, the service UAV, U_j^S , is offering S_2 . We even have the discounting factors of δ_i^T and δ_j^S for task and service UAVs, respectively.

We represent a player's payoff as $P(\cdot)$. Assume that the minimum and maximum payoff U_i^T is denoted as $\underline{V}$ and $\bar{V}$, respectively.

$$P(U_i^T)_{min} = \underline{V} \quad (19a)$$

$$P(U_j^S)_{max} = \bar{V} \quad (19b)$$

U_j^S makes an offer: Let, after a history of h , the service UAV makes an offer. In the offer, U_j^S offers $(a_2, 1 - a_2)$, i.e. a_2 to U_i^T and $(1 - a_2)$ to itself.

$$P(U_i^T)^S = a_2 \quad (20a)$$

$$P(U_j^S)^S = 1 - a_2 \quad (20b)$$

However, from Eq. (19), we have the best and worst payoff U_i^T can receive and considering it with the discounting factor,

$$U_i^T \text{ will accept if } a_2 \geq \delta_i^T(\bar{V}) \quad (21a)$$

$$U_i^T \text{ will reject if } a_2 \leq \delta_i^T(\underline{V}) \quad (21b)$$

Therefore, from Eq. (21), we conclude that:

$$U_i^S \text{ can at least retain itself } 1 - \delta_i^T(\bar{V}) \quad (22a)$$

$$U_i^S \text{ can at most retain itself } 1 - \delta_i^T(\underline{V}) \quad (22b)$$

U_i^T makes an offer: Now, let the task UAV is making an offer. U_i^T offers $(a_1, 1 - a_1)$ in this offer, i.e., a_1 to itself and $(1 - a_1)$ to U_j^S .

$$P(U_i^T)^T = a_1 \quad (23a)$$

$$P(U_j^S)^T = 1 - a_1 \quad (23b)$$

From Eq. (23), the service UAV knows the least and most payoff it can get. So, U_j^S accepts if it is offered more than or equal to the least payoff or less than or equal to the most payoff, i.e., U_j^S accepts depending upon the following conditions:

$$1 - a_1 \geq \delta_j^S(1 - \delta_i^T(\bar{V})) \quad (24a)$$

$$1 - a_1 \leq \delta_j^S(1 - \delta_i^T(\underline{V})) \quad (24b)$$

where $\delta_j^S(1 - \delta_i^T(\bar{V}))$ and $\delta_j^S(1 - \delta_i^T(\underline{V}))$ are the minimum and maximum payoffs of U_j^S respectively, in other words, U_i^T can retain to itself a maximum payoff in case of Eq. (24a) and a minimum payoff in case of Eq. (24b). Additionally, since a_1 is the payoff U_i^T can retain to itself, mathematically,

$$1 - \bar{V} \geq \delta_j^S(1 - \delta_i^T(\bar{V})) \quad (25a)$$

$$1 - \underline{V} \leq \delta_j^S(1 - \delta_i^T(\underline{V})) \quad (25b)$$

Solving Eqs. (25a) and (25b), we obtain:

$$\underline{V} \geq \frac{(1 - \delta_j^S)}{(1 - \delta_i^T \delta_j^S)} \quad (26a)$$

$$\bar{V} \leq \frac{(1 - \delta_j^S)}{(1 - \delta_i^T \delta_j^S)} \quad (26b)$$

Further, for SPNE, $\underline{V} = \bar{V} = V$. Therefore, from Eq. (26) we conclude:

$$V = \frac{(1 - \delta_j^S)}{(1 - \delta_i^T \delta_j^S)} \quad (27)$$

U_i^T should offer $(V, 1 - V)$ to attain equilibrium. Therefore, the equilibrium payoffs of U_i^T and U_j^S are $\frac{(1 - \delta_j^S)}{(1 - \delta_i^T \delta_j^S)}$ and $1 - \frac{(1 - \delta_j^S)}{(1 - \delta_i^T \delta_j^S)}$, respectively. $\square$

The equilibrium payoff values, as derived in Eq. (27), show that the final outcome depends on the discounting factors δ_i^T and δ_j^S . A higher discount factor means the player values future payoffs more, leading to a slower but still guaranteed convergence. In practice, after a few rounds of negotiation, the offers reach a steady state, confirming that the game settles into a predictable and fair division of utility. Therefore, the iterative nature of the bargaining process ensures that both UAVs eventually reach an agreement that neither has an incentive to deviate from, demonstrating the robustness of our offloading strategy.

From Eq. (27), the award $\mathcal{A}_{ij}^t$ is given as:

$$\mathcal{A}_{ij}^t = \frac{(1 - \delta_j^S)}{(1 - \delta_i^T \delta_j^S)} \quad (28)$$

Algorithm 1 describes the process for selecting service UAVs for task offloading. It takes as input the set of all task assignments ($\mathbb{T}$) and the set of all service UAVs (U^S) and outputs the set of assigned service UAVs (U_{ass}^S). For each task, the algorithm evaluates each service UAV's eligibility and computes trust between task and service UAV. The UAV with the highest trust score not already assigned is selected for the task.

Algorithm 1 Service UAV selection in TOFU

INPUTS:

- 1: $\mathbb{T} = \{\mathbb{T}_1, \mathbb{T}_2, \mathbb{T}_3, \dots, \mathbb{T}_k\}$ $\triangleright$ set of all task assignments
- 2: U^S $\triangleright$ Set of all service UAVs

OUTPUT: U_{ass}^S

$\triangleright$ Set of assigned service UAVs

PROCEDURE:

- 1: Initialize an empty set of U_{ass}^S
- 2: **for** $i \leftarrow 0$ to k **do**
- 3: $\zeta = 0$
- 4: **for** $j \leftarrow 0$ to $n(U^S)$ **do**
- 5: Compute $e(U_j^S, t_i, U_i^T)$ using Eq. (2)
- 6: **if** $e(U_j^S, t_i, U_i^T) = 1$ **then**
- 7: Compute $\mathcal{R}_j$ using Eq. (8) or (10) (If it is the first trial)
- 8: Compute $\mathcal{D}_{ij}$ using Eq. (14)
- 9: Compute ζ_{ij} using Eq. (1)
- 10: **if** $\zeta_{ij} > \zeta$ and $U_j^S \notin U_{ass}^S[i]$ **then**
- 11: $\zeta \leftarrow \zeta_{ij}$
- 12: $U_{ass}^S[i] \leftarrow U_j^S$
- 13: **end if**
- 14: **end if**
- 15: **end for**
- 16: **end for**
- 17: **return** U_{ass}^S

4.2.3 Fallback Mechanism

Task offloading in a dynamic and resource-constrained UAV environment may not always succeed. Unlike DeTO, DPTO, and SOA, which do not implement a fallback mechanism, TOFU accounts for potential offloading failures. When a task is offloaded to the most trusted UAV, but the transmission fails due to communication errors, network congestion, or UAV mobility, TOFU dynamically selects the next most trusted UAV for execution. This fallback mechanism ensures that task execution continues within the allowed latency constraints. If the retransmission and execution exceed the maximum permissible delay, the task is considered failed, and the reputation of the failing UAV is adjusted accordingly. This approach significantly enhances the reliability of task offloading in UAV-as-a-Service platform.

5 PERFORMANCE ANALYSIS

5.1 Simulation Design

This section discusses the performance of the proposed scheme, TOFU. We perform simulation for TOFU using Algorithm 1. We first explain the simulation design and then discuss the result in detail. In the simulation, we consider a simulation area of $10\text{km} \times 10\text{km}$, where 5 to 20 heterogeneous UAVs are present. We execute the simulation for 5 to 100 trials for each of the results, and each trial has a certain number of tasks. An end-user registers with the UaaS platform and pays the charges based on the usage of the resources. In the process, a task UAV offloads a task to a service UAV based on trust. Unlike the traditional Service-Oriented Architecture, the TOFU is different and unique because we consider trust in this approach, which is not considered in the existing literature. However, we attempted to compare the performance of TOFU with the

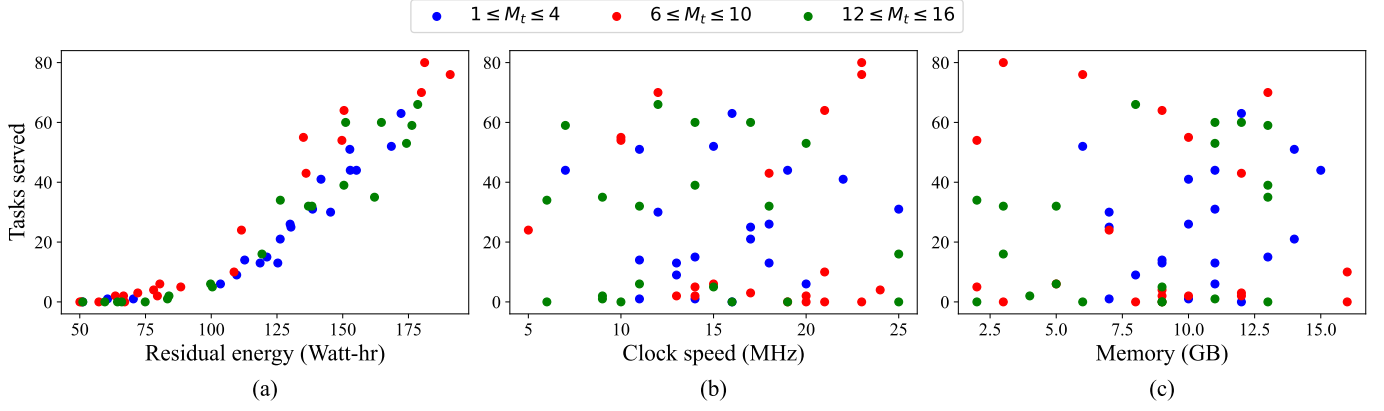


Fig. 2: Number of tasks served vs computational resources for tasks with different memory requirement ranges.

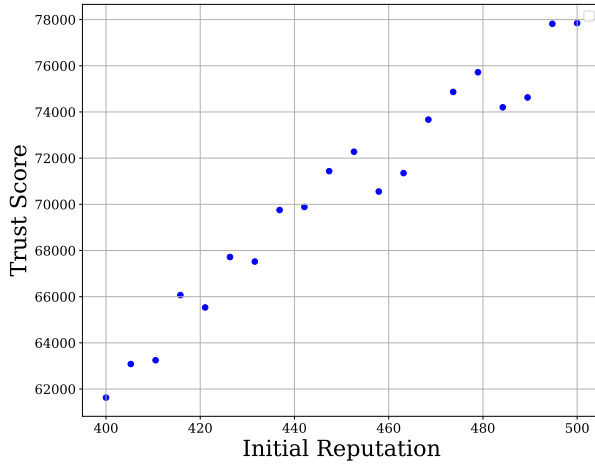


Fig. 3: Variation of Trust score w.r.t initial reputation.

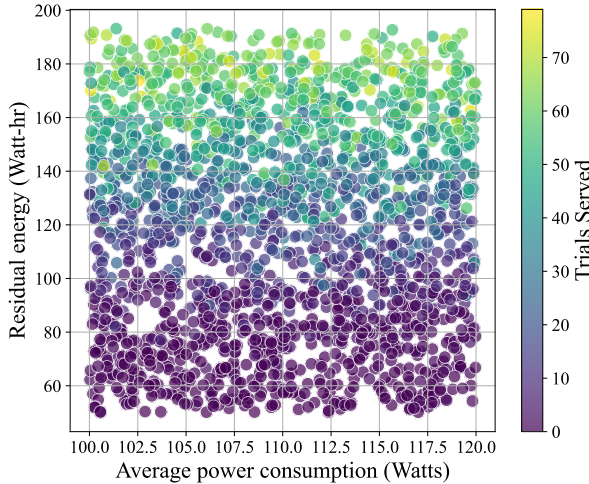


Fig. 4: Variation in trials served with residual energy and power consumption.

three most relevant research works as benchmark schemes. The simulation parameters are shown in Table 2.

Benchmark Schemes

(i) DeTTO: Dependency-aware trustworthy task offloading

in vehicular IoT transfer [19]. In this work, the authors formulate the task offloading problem as a graph optimization problem, where the authors aim to minimize only transmission energy and delay. The authors categorize tasks into three different types: (i) fully independent, (ii) fully dependent, and (iii) partially dependent tasks. For fully independent tasks, authors proposed *greedy algorithm* to determine an optimal set of nodes for task offloading.

(ii) DPTO: A Deadline and priority-aware task offloading in fog computing framework leveraging multilevel feedback queueing [31]. In this work, the authors designed a novel delay-dependent priority-aware task offloading strategy for scheduling and processing tasks. The proposed strategy assigns priority to each task and then assigns it to a suitable multilevel queue. The DPTO strategy selects an optimal computing device based on its resource availability and transmission time from the device.

(iii) SOA: A Strategic Game for Task Offloading among Capacitated UAV-Mounted Cloudlets [32]. This work addresses the task offloading problem in a network of UAV-mounted cloudlets, where UAVs act as edge computing nodes for IoT devices. Considering the limited computational resources and capacity, UAVs must balance energy consumption while ensuring service quality. The problem is formulated as an integer program and modeled as a strategic offloading game. The authors prove the existence of a pure-strategy Nash equilibrium and propose an algorithm to achieve it.

5.2 Complexity and Convergence Analysis

To evaluate the efficiency of the proposed algorithm, we analyze its computational complexity and convergence properties. The algorithm iterates over k tasks, and for each task, it evaluates $n(U^S)$ service UAVs. Within this nested loop, the algorithm performs several operations, including computing eligibility using Eq. (2), reputation score using Eq. (8) or Eq. (10) for first-time computations, data transfer rate using Eq. (14), and trust computation using Eq. (1). These operations involve basic arithmetic and conditional checks, leading to a per-iteration complexity of $\mathcal{O}(1)$. Given the two-level looping structure, the overall computational complexity of the algorithm is $\mathcal{O}(k \cdot n(U^S))$. In practical settings, where k (number of tasks) and $n(U^S)$ (number

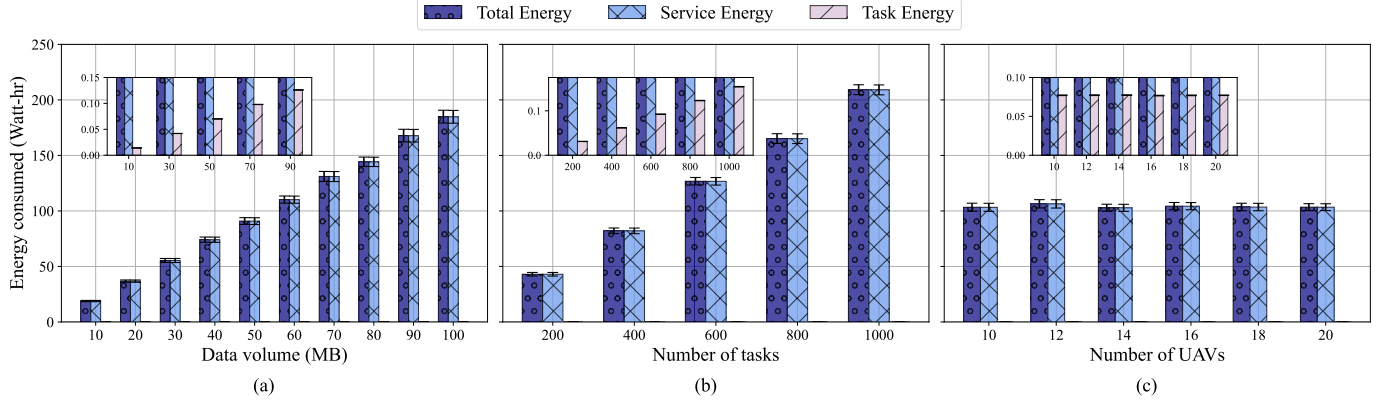


Fig. 5: Variation of energy consumption with the data volume of the task, number of tasks, and number of UAVs.

TABLE 2: Simulation parameters.

Parameter	Value
Simulation area	10km × 10km
Number of UAVs	5 - 20
Number of trials	100
Number of tasks per trial	2 - 10
Data volume of task	10 - 1000 MB
Clock speed of UAV	5 - 25 MHz
Memory of UAV	2 - 16 GB
Initial energy	50 - 200 Watt-hr
Power consumption	100 - 120 Watts
Instructions per unit data	1 - 2 Million
Tolerable latency of task	1 - 10 ms
Bandwidth	10 MHz
Path loss coefficient	2.5 - 4 dB
Transmission power	0.1 - 1 W
Noise power	10 ⁻⁹ W

of service UAVs) remain within a manageable range, the algorithm demonstrates efficient execution. Moreover, the TOFU algorithm is designed to be scalable with the number of UAVs and tasks. Trust computations are localized and updated only between interacting UAVs, significantly reducing global synchronization overhead. The reputation updates follow a debit-credit mechanism, which requires only minimal information exchange between the task and service UAVs. As such, the communication overhead per offloading event remains constant and does not grow with the total number of UAVs in the network.

For each task, TOFU first checks the eligibility of available UAVs using simple arithmetic comparisons (Eq. 2). For eligible UAVs, it computes reputation and trust scores through weighted sums and conditional updates (Eq. 8, 10, 1). The data transfer rate is calculated using the Shannon capacity formula (Eq. 14), which involves logarithmic and basic arithmetic operations. These computations are performed across all service UAVs ($n(U^S)$) for every task (k). Additionally, TOFU applies local trust updates and filters out ineligible UAVs early, which helps avoid unnecessary processing. As a result, the algorithm maintains minimal computational overhead, making it suitable for real-time

task offloading in UaaS environments.

For convergence analysis, we consider the iterative process of selecting the most trustworthy UAV for task execution. The algorithm ensures that at each step, the UAV with the highest trust score is assigned, thereby improving task execution reliability. Since the trust score ζ_{ij} is updated iteratively based on the UAV's reputation, data transfer rate, and past interactions, the trust model exhibits a non-decreasing trend. Furthermore, as UAVs with lower trust scores are progressively filtered out, the selection process stabilizes after a finite number of iterations. Thus, the proposed algorithm converges in a deterministic manner, ensuring reliable UAV-task assignments without infinite loops or oscillations.

5.3 Result and Discussions

In this section, we discuss the performance analysis of the proposed task offloading scheme considering various factors such as energy consumed, the data volume of the task, resources available at the device, the reputation of the device, and delay incurred in executing the tasks.

One of the objectives of the proposed model is to select a UAV to offload the task which is suitable for serving for a longer time. Fig. 2 presents a comparative analysis of the number of tasks served by UAVs with respect to different resource constraints. In Fig. 2a, we observe that UAVs with higher residual energy consistently serve more tasks than those with lower residual energy, irrespective of memory requirements. This aligns with the expectation that UAVs with greater energy availability can sustain longer operation times, making them more suitable for task offloading. In contrast, Fig. 2b and Fig. 2c indicate that there is no clear correlation between the number of tasks served and the UAV's clock speed or memory capacity. This is because clock speed and memory do not directly impact the UAV's reputation, except in the initial stages when these factors primarily determine reputation. Furthermore, we observe that as task memory requirements increase, the number of UAVs capable of serving tasks decreases. This is expected, as only a subset of UAVs possesses sufficient memory to handle tasks with higher memory demands, thereby reducing the pool of available UAVs for offloading. Even when tasks are independent of clock speed and memory, the trust

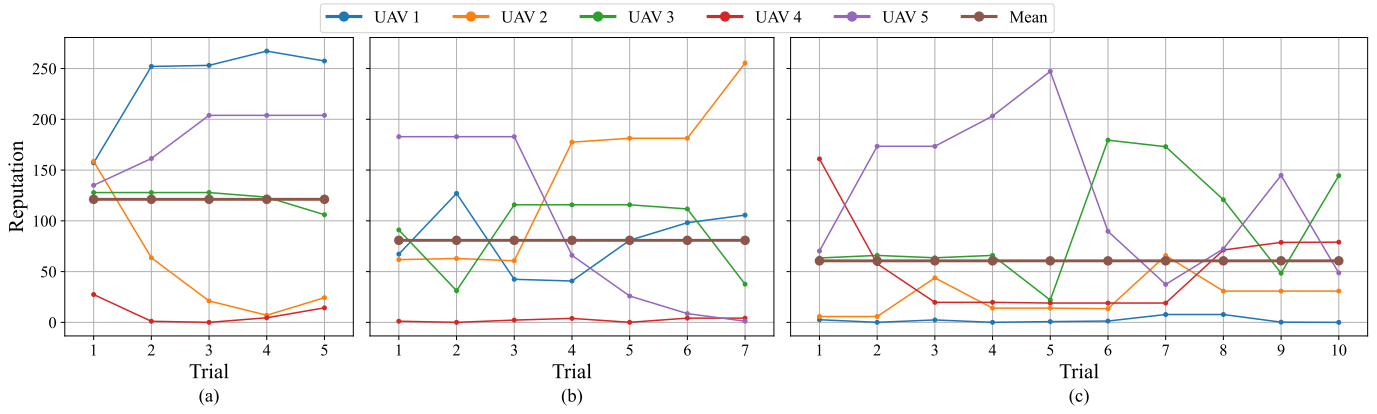


Fig. 6: Variation of reputation of UAVs with trials.

score of a UAV is still influenced by the product of these two parameters. Fig. 3 illustrates the relationship between initial reputation, the product of clock speed and memory, and trust score. It is observed that the trust varies linearly with the initial reputation of the UAV. This reputation plays a critical role during bootstrapping, influencing early trust formation and guiding task offloading decisions toward UAVs with higher computational capabilities. Fig. 4 illustrates the number of trials served by each UAV with respect to their residual energy and power consumption over 100 iterations. The trend observed here aligns with the findings in Fig. 2, reinforcing that a UAV's ability to serve tasks is primarily influenced by its residual energy. Regardless of a UAV's power consumption, higher residual energy increases the likelihood of participating in task execution, enabling sustained operations for a longer duration without exhausting resources.

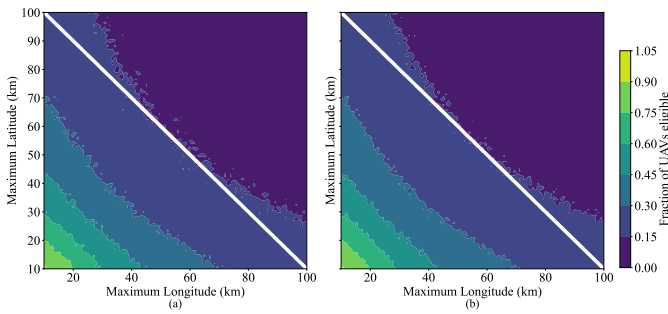


Fig. 7: Fraction of UAVs eligible across different ranges of service area dimensions.

We compare the energy consumption of the UAVs with various factors such as data volume of the task, number of tasks, and number of UAVs in the system in Fig. 5. In the plot, we represent the energy consumed by task UAV as task energy, the energy consumed by service UAV as service energy, and the total energy consumed as total energy. From the plot, we infer – (i) task energy is almost negligible compared to total energy and service energy, (ii) energy consumption increases with an increase in the data volume of the task and number of tasks, and (iii) energy consumption does not vary significantly with change in the number of UAVs in the system. Since larger tasks take longer

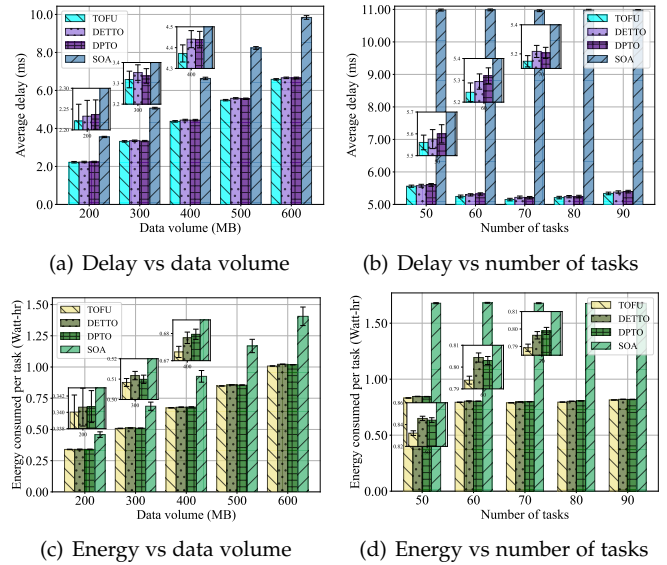


Fig. 8: Comparison of delay and energy consumption for different offloading techniques.

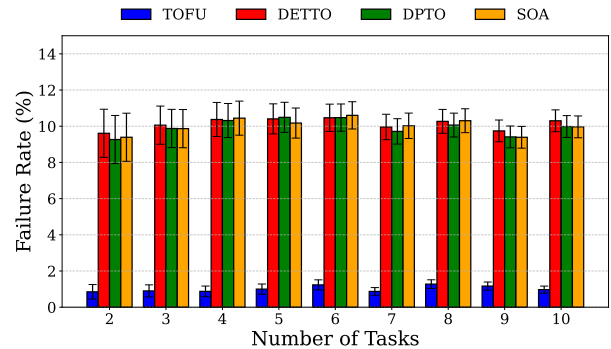


Fig. 9: Comparison of failure rate with increase in number of tasks for different offloading techniques.

to transmit and execute, they consume more energy than those with smaller sizes. Similarly, a higher number of tasks require more number of packet transmissions, and energy consumption also increases. However, an increase in the

TABLE 3: Theoretical comparison of TOFU with existing schemes

Feature	TOFU	DeTTO [19]	DPTO [31]	SOA [32]
Trust-Based Task Offloading	✓	✓	×	×
Deadline-Aware Scheduling	✓	×	✓	×
Real-Time Adaptability	✓	×	×	✓
Dynamic Trust Evaluation	✓	×	×	×
Fallback mechanism	✓	×	×	×
Incentivized Offloading Mechanism	✓	×	×	✓
Energy Efficiency	✓	×	×	×
Computational Efficiency in Trust Evaluation	✓	×	×	×
Scalability in Dynamic UAV Networks	✓	×	×	✓

number of UAVs does not account for the energy consumed. Therefore, if the number and nature of the tasks remain constant, the energy consumed also remains constant.

Fig. 6 illustrates the variation in the reputation of each UAV across different trials, along with their mean reputation values. While individual UAV reputations fluctuate without following a specific pattern, the mean reputation remains constant regardless of the trial or the total number of trials. This stability in the mean reputation suggests that while UAV reputations change due to dynamic interactions and performance variations, the overall system maintains equilibrium, reinforcing the validity of Property 1. Fig 7 depicts the variation of the fraction of eligible UAVs to serve the tasks across different maximum limits of service area dimensions. We also notice that when the service area is the smallest, the probability of eligibility for a UAV is almost equal to 1. However, as the area increases, the probability eventually decreases and falls to 0. The possible reason behind such a trend is that – as the area expands, UAVs disperse over a larger region, resulting in a sparse network that leads to lower data rates.

We also compare the delay and energy consumption of the proposed model with existing offloading techniques, DeTTO, DPTO, and SOA. Figs. 8a and 8b depict the comparison of delay for the data volume of the task and the number of tasks. We notice that the delay increases with an increase in data volume, but no pattern is observed in the case of varying numbers of tasks; regardless, TOFU offers service with a lower delay compared to the other techniques. This is because in DPTO, the task is offloaded to a device that has enough resources to execute the task within the deadline but doesn't account for the best resources. DeTTO considers only the energy consumption incurred in task offloading and does not consider the energy incurred in executing the task as it was proposed for vehicular IoT. SOA, on the other hand, prioritizes system-wide energy minimization through Nash equilibrium strategies, resulting in higher delay compared to TOFU. Although SOA minimizes energy theoretically, it considers only static tasks and resources, leading to suboptimal energy efficiency. The aforementioned reasons also support Fig. 8c and 8d where a similar trend is observed. While DeTTO, DPTO, and SOA have addressed task offloading challenges in vehicular IoT and fog computing, respectively, their applicability to the UaaS platform remains limited.

Fig. 9 illustrates the comparison of failure rates between DeTTO, DPTO, SOA, and the proposed TOFU scheme. It is observed that TOFU consistently exhibits a lower failure rate across various scenarios. This improvement is attributed to the fallback mechanism of the TOFU mentioned

in 4.2.3. Despite the presence of a fallback mechanism, TOFU still encounters occasional task failures, primarily due to timeouts when alternative trustworthy UAVs are either unavailable or unable to meet the task's deadline constraints during reassignment. Furthermore, as shown in Table 4, TOFU achieves the highest fraction of participating UAVs among all compared techniques. This is attributed to its incentivized offloading strategy and reputation-aware bargaining model, which actively promote the inclusion of UAVs with lower reputations by enabling them to negotiate for higher incentives. As a result, TOFU encourages broader participation and resource utilization across the network.

TABLE 4: Fraction of UAVs participating in the offloading.

Number of UAVs	TOFU	DeTTO [19]	DPTO [31]	SOA [32]
5	0.94	0.49	0.84	0.48
10	0.82	0.36	0.74	0.35
20	0.73	0.25	0.65	0.28

6 CONCLUSION

In this work, we proposed a scheme, Trust-based Task Offloading for UaaS (TOFU). We adopted Rubinstein's bargaining model with fixed discounting factors to model the dynamics of UAVs in UaaS. In the proposed scheme, task UAV, which is willing to offload the task, is the first one to make an offer to give a fraction of its reputation, and the service UAV to which it is offloading is ready to execute the offloaded task and receives that reputation. We also demonstrated the existence of a Sub-game-Perfect Nash Equilibrium that ensures the best payoff for both the UAVs. Through simulations, we observed that TOFU outperforms benchmark schemes – DETTO, DPTO, and SOA in terms of both delay and energy consumed by the system for a given task. In the future, we aim to explore integrating blockchain technology to enhance the security and transparency of task offloading in UaaS. Leveraging blockchain technology for task offloading makes the system more robust and fault-tolerant due to its decentralized nature. Expanding the model to incorporate these security measures will enhance trust among UAVs and provide a more resilient and secure scheme for UAV-based applications in diverse real-world scenarios.

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